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Photochemical Reactions of Tris(oxalato)Iron (III)

A first-year chemistry experiment

The increasing cost of dwindling energy sources has made the majority of first-year chemistry students cognizant of the potential of solar or light energy. The student's formal introduction to the harvesting of this energy, however, is often through the study of photosynthesis. Unfortunately, the complexity of this marvelously intricate series of coupled photochemical and thermal reactions can overshadow the basic and fundamental concepts of a photoinduced reaction. Consequently, few students have a clear appreciation of the energy content of light or its potential to induce a chemical reaction. To enhance this awareness, the qualitative and quantitative ideas of photochemistry have been introduced for the past three years at Queens College through an experiment based on the well characterized photoreduction of the standard actinometric reagent potassium ferrioxalate, $K_3Fe(C_2O_4)_3$.²

The experiment consists of a mix of qualitative studies based on visual observations including the production of a blueprint, and more quantitative kinetic aspects based on spectral measurements. The qualitative parts alone may be useful for large enrollment non-majors courses where spectrophotometers or colorimeters may not be available, while for a majors or pre-medical group a blend of qualitative and quantitative exercises is more appropriate. The most able students in our classes complete all the exercises below in a 3-hr laboratory class; however, we usually suggest that the more detailed kinetics exercise (described in Section II of "Experimental") be left until the end since it may be necessary for slower students to omit it, and this can be done without greatly diminishing the value of the experiment as a whole.

The quantitative parts of the experiment require an application of Beer's Law and also some introduction to elementary photochemical kinetics. The students at Queens College are assigned appropriate explanatory material and do a pre-laboratory exercise to orient them to these topics. Copies of the instructions issued to students may be obtained from the authors.

The experiment itself involves absorption of light of ≤ 450 nm by potassium ferrioxalate. This induces an efficient intramolecular electron transfer resulting in the oxidation of oxalate, ultimately to carbon dioxide, and the reduction of iron (III) to iron (II). Conventionally, the photochemical reaction is quantitated by complexation of the Fe^{2+} with 1,10-phenanthroline.² However, the time required to develop the Fe^{2+} -phen color is a significant drawback for use in a relatively short laboratory class. To circumvent this time-consuming step, we have instead detected the Fe^{2+} by adding potassium ferricyanide, $K_3Fe(CN)_6$ to form the deeply blue ferroferricyanide complex referred to as Turnbull's blue.⁴ Since the blue color develops within the time of mixing, the student is able,

in a 3-hr laboratory period, to verify that the reaction is induced by blue light, to measure the intensity dependence of the reaction, and to use the photochemical reaction to make a "blueprint."

The cost per set-up is modest; about \$20 covers the price of light source and red filter required for the experiment.

Experimental

A. Preparation of $Fe(C_2O_4)_3^{3-}$

CAUTION: The reagents used in this experiment, especially the $Fe(CN)_6^{3-}$ solution are poisonous and must be handled with care. Either a graduated cylinder, or a pipet with bulb should be used to dispense measured volumes.

The ferrioxalate is prepared *in situ* by dissolving 0.6 g of $Fe(NO_3)_3 \cdot 9H_2O$ in a 250 ml Erlenmeyer flask containing 100 ml of distilled water, followed by the addition of 0.4 g of oxalic acid. Formation of ferrioxalate is indicated by the appearance of a yellow color. This yellow solution is sensitive to light; direct exposure to sunlight should be avoided and the solution should be stored in a dark place, such as the student's drawer, when not in use. Inevitably the logistics of a freshman laboratory will demand that actual experimental manipulations of these solutions be done in room lights, but this causes no major problems as long as the students work reasonably quickly.

B. Photochemical Reaction of $Fe(C_2O_4)_3^{3-}$

The light sources used in the experiment are 300 W Westinghouse reflector type flood lamps. To minimize the students' direct exposure to these very bright lamps, the lamps are housed in one gallon metal solvent cans. The bottom of the can is removed and a standard ceramic lamp socket is bolted to the inner surface of the top of the can. The wires to the socket are passed through the mouth of the can and a metal rod is bolted through a hole drilled in the side of the can so that the light sources can be mounted easily on a ring stand.

Since these lamps generate a considerable amount of heat, **the students should be forewarned not to touch the lamp housing and to turn the lamp on only when needed for the irradiation.**

To carry out the photochemical reaction, 10 ml of the $Fe(C_2O_4)_3^{3-}$ solution is added to each of four numbered 1.8 $\times$ 15 cm test tubes. One tube is kept away from the light source and irradiated for 10, 20, and 30 sec, respectively. If lower wattage lamps are used, longer irradiations may be necessary.

The amount of Fe^{2+} formed in the photochemical reaction is determined by the addition of 1 ml of a 0.3 M $K_3Fe(CN)_6$ solution to each of the irradiated solutions and the reference solution. The reference solution remains pale yellow, but the irradiated solutions develop the blue color of the ferroferricyanide complex. A visual comparison of the irradiated solutions against a white background shows that the intensity of the blue color is proportional to the time of irradiation. Under conditions of constant intensity, the number of Fe^{2+} ions formed is proportional to the number of photons absorbed which, in turn, is proportional to the time of exposure.

Although the visual comparison illustrates the relationship between the time of exposure and the extent of reaction, the experiment is more quantitative and versatile if the absorbance of the ferroferricyanide solution is measured at 700 nm with a spectrophotometer. From the ratio of absorbance of the irradiated solutions to the absorbance of a reference solution prepared by adding 1 ml of 0.03 M $K_3Fe(CN)_6$ to 10 ml of $Fe(C_2O_4)_3^{3-}$ solution containing 1 ml of 10^{-2} M ferrous ammonium sulfate in 10^{-5} M sulfuric acid, the concentration of Fe^{2+} is calculated. The rate of the photochemical reaction is then calculated from the slope of a plot of the concentration of Fe^{2+} formed in the photolysis versus the irradiation time.

Since these lamps produce a considerable amount of heat, the more skeptical student may question whether the reaction was induced by

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² Parker, C. A., *Proc. Roy. Soc. (London)*, A220, 104 (1953); Hatfield, C. G., and Parker, C. A., *Proc. Roy. Soc. (London)*, A235, 518 (1956).

³ Balzani, V., and Carassiti, V., "Photochemistry of Coordination Compounds," Academic Press, 1970.

⁴ Cotton, F. A., and Wilkinson, G. W., "Advanced Inorganic Chemistry," 3rd Ed., Interscience, 1977.

⁵ In view of the toxicity of this reagent, this solution should be prepared by the instructor prior to the laboratory.

light. To establish that the reaction was not due to heat, a test tube containing 10 ml of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ is placed in a beaker of boiling water for about 1 min. Then 1 ml of 0.03 M $\text{K}_3\text{Fe}(\text{CN})_6$ is added. The absence of an appreciable blue color establishes that the reduction of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ is due only to light.

In the above experiments, the concentration of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ solution is sufficient to absorb essentially all of the blue and uv light emitted by these lamps. Under these conditions of high optical density, the rate of a photochemical reaction is dependent only on the light intensity and independent of the concentration of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$.³ Because the light source is polychromatic and more concentrated solutions of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ absorb a larger fraction of the visible light emitted by these lamps, rigorously establishing that the rate of the photochemical reaction is independent of the concentration of the photoactive substrate is not possible. If a spectrophotometer is available, however, the first-order dependence of the photolysis rate can be verified.

A general rate expression for the reaction can be written as

$$\frac{d(\text{Fe}^{2+})}{dt} = \phi I_a^n \quad (1)$$

where ϕ is the quantum efficiency of the reaction and I_a is the absorbed intensity. Under these conditions of high optical density, I_a equals the intensity incident on the sample, I_o . If we assume that these lamps are unfocused point sources of constant intensity I , the irradiance at the $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ solution surface, I_o , varies inversely as the square of the distance between the source and the sample.⁶ Equation (1) can then be written as

$$\frac{d(\text{Fe}^{2+})}{dt} = \phi(I/r^2)^n \quad (2)$$

where r is the distance between the light source and the sample. From the log form of eqn. (2)

$$\log \frac{d(\text{Fe}^{2+})}{dt} = (\log \phi + n \log I) + n \log \frac{1}{r^2} \quad (3)$$

the intensity dependence can be extracted from a plot of $\log d(\text{Fe}^{2+})/dt$ versus $\log (1/r^2)$.

To determine experimentally the intensity dependence, n , 10 ml of the $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ solution is added to each of three numbered test tubes. To help in positioning each of these 10 ml aliquots, a chalk line drawn on the bench top normal to the center of the lamp face is recommended. Along this line, one of the 10 ml aliquots is placed in a wire test tube rack 3 in. from the face of the lamp and irradiated for 30 sec. The rack is then moved along the line to a position 6 in. from the light source and another 10 ml aliquot is irradiated for 30 sec. With the third aliquot, the experiment is repeated at a position 9 in. from the lamp. To each of the irradiated solutions 1 ml of 0.03 M $\text{Fe}(\text{CN})_6$ is added and the absorbance of the reference solution containing a known amount of Fe^{2+} described above, the rate of the photochemical reaction is calculated at each position. The log of these rates is plotted versus the log of $1/r^2$ and the value of n obtained from the slope of the plot.

The accuracy and precision associated with the determination of n vary from group to group of students. For a spectrum of the better students, values ranging from $n = 1.2$ – 1.6 have been obtained. One possible source of problems is the stability with time of the flood lamps employed in the experiments. While these normally appear to be quite stable, and satisfactory results are generated, we occasionally encounter poorer results during times of line voltage fluctuations, e.g. on hot summer days. Prolonged exposure to room lights will of course also tend to lead to high estimated for n .

C. The Absorption Spectrum of Ferrioxalate Ion and the Wavelength Dependence of the Photochemical Reaction

Although these lamps emit light ranging from the relatively low energy infrared to the higher energy blue and uv light, only the latter is absorbed by the $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ and is of sufficient energy to induce the photochemical reaction. The frequencies at which light is absorbed is obtained from the absorption spectrum of the $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$. Using a colorimeter and a test tube filled with water as a "blank," the absorbance of the $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ solution is measured at 25-nm intervals over the 700-nm to 400-nm (or lower) range. From a plot of absorbance versus wavelength, it is apparent that red light, $\lambda \geq 600$ -nm, is not absorbed by the complex. Since light must be absorbed by the complex to initiate the photochemical reaction, red light should not induce any reaction. To confirm this, a red filter (Carolina Biological Supply, Burlington, N.C. 27215, filter number 68-6700) is used to isolate light of $\lambda \geq 650$ -nm.

To each of the three numbered test tubes, 10 ml of the $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ solution is added. One tube is set aside as a reference. Another tube is irradiated with white light for 20 sec. Holding the red filter so that no portion of the third aliquot of $\text{Fe}(\text{C}_2\text{O}_4)_3^{3-}$ is directly exposed to the lamp, the third solution is irradiated for the same time. On adding 1 ml of 0.03 M $\text{K}_3\text{Fe}(\text{CN})_6$, the aliquot exposed to white light develops an intense blue color while the reference and the aliquot exposed to red light remain pale yellow.

D. Applications of Photochemical Reactions.

Photosensitive reactions are the basis of the commercial fields of photography and photocopying. One such commercial use of a photochemical reaction, although now largely displaced by newer techniques, is blueprinting. Blueprinting consists of treating paper with a ferrioxalate solution to make it light sensitive. An opaque object or diagram is placed on top of the photosensitive paper and both are exposed to light. Light penetrates the nonopaque regions of the original and induces a photochemical reaction on the underlying photosensitive paper. Exposure is prevented by the opaque regions of the original, and no photochemical reaction occurs. On development with $\text{K}_3\text{Fe}(\text{CN})_6$, the exposed region appears light blue, while the unexposed regions remain white.

A ferrioxalate solution is prepared by dissolving 1 g of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 1 g of oxalic acid in 25 ml of water. Ten to fifteen milliliters of this solution is placed in a Petri dish and used to soak thoroughly 8 cm diameter pieces of filter or white paper. With tongs or tweezers, the paper is removed and the excess reagent allowed to drip into the dish. Treated paper is dried away from bright lights.

The dry treated paper is laid flat about 3 in. below the lamp, and coins, keys, or other available flat opaque objects are placed on the paper. After irradiating the paper for about 2 min, the paper is dipped into 25 ml of 0.03 M $\text{K}_3\text{Fe}(\text{CN})_6$. The exposed regions of the paper turn blue while the region under the opaque objects remains white. After washing off the excess reagent by dipping the paper in a Petri dish filled with distilled water, the paper is dried and the "blueprint" submitted in the laboratory report.

⁶ If the angle between the incident ray and the normal to the surface is θ , the intensity at the surface is also dependent upon $\cos \theta$. Neglecting this dependence, however, introduces an error which is generally less than the variation in the data.